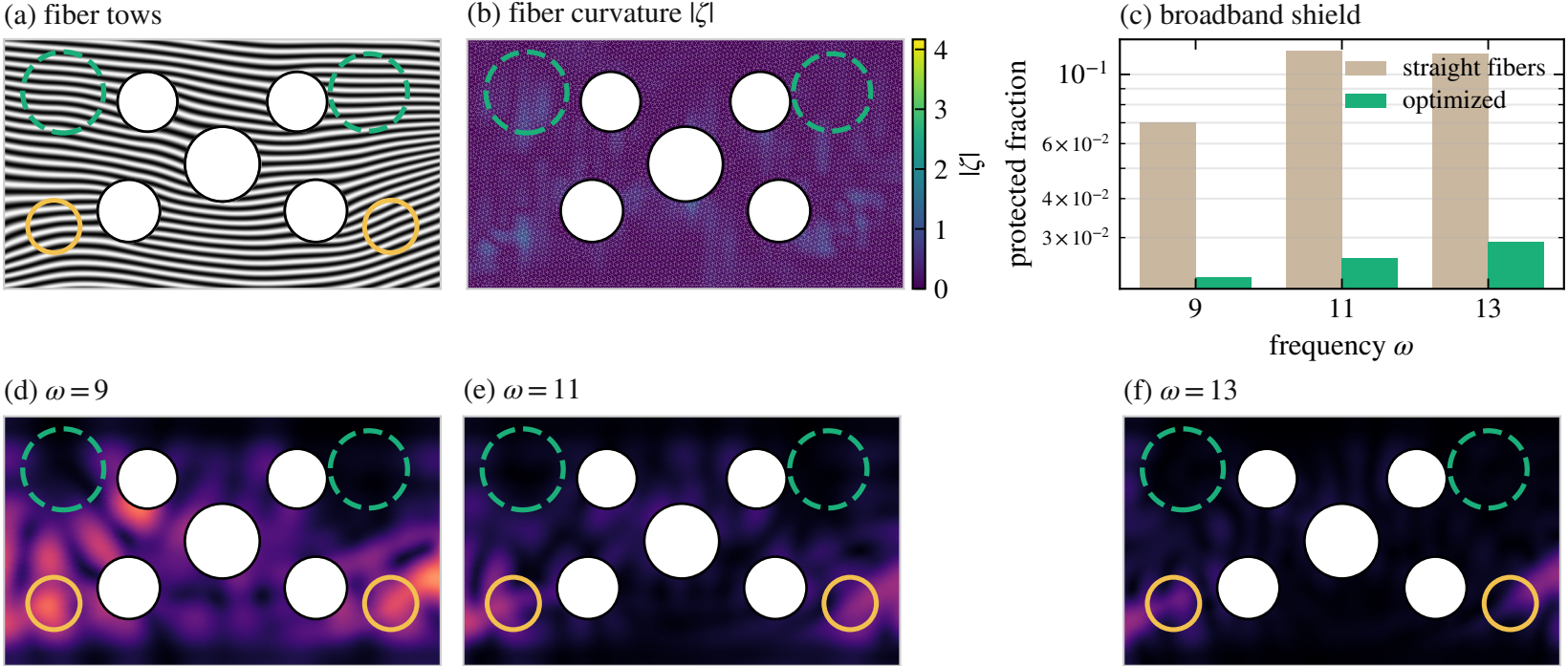

(a) fiber tows
(b) fiber curvature |ζ|
|ζ|
4
3
2
1
0
(c) broadband shield
protected fraction
10^{-1}
6×10^{-2}
4×10^{-2}
3×10^{-2}
straight fibers
optimized
9
11
13
frequency ω
(d) ω = 9
(e) ω = 11
(f) ω = 13